

Pediatric airway management remains a critical challenge due to the anatomical variability observed in the 0–2 age group, where rapid tracheal development significantly impacts intubation procedures

Faculty Advisor:	[Name] : K. Banu Köse [Faculty] : BME [E-mail] : kbkose@medipol.edu.tr [Phone] : 5446
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Industry Partner	-
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Overview of the Project:

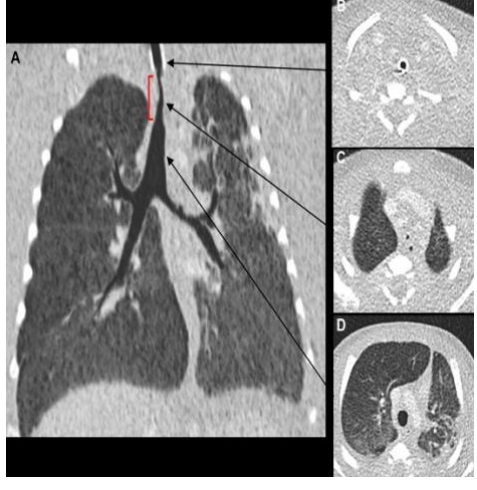
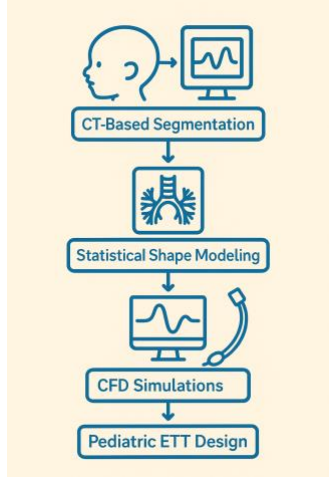
Pediatric airway management in the 0–2 age group remains a complex challenge due to the high degree of anatomical variability. This project proposes a digital twin framework to systematically capture and analyze this variability. Retrospective CT datasets will be preprocessed and segmented, after which implicit neural representations (INR) will be employed to encode the pediatric trachea as a continuous function rather than a discrete mesh or voxel model. This approach enables precise and compact representation of geometry while preserving subtle morphological details. Statistical shape analysis (SSA) will then be applied across the patient cohort to quantify tracheal variability, derive an average shape model, and identify dominant deformation modes. These statistically grounded models will serve as the basis for air flow dynamics analysis. Ultimately, the project seeks to establish a validated digital-twin-based pipeline that bridges anatomical variability with functional performance to address one of the most pressing unmet needs in pediatric airway management.

Objective of the Project:

The primary objective of this project is to establish a digital twin-based framework for pediatric tracheal modeling in the 0–2 age group by integrating statistical shape analysis with implicit neural representations. This framework aims to quantify anatomical variability, construct an average tracheal model, and evaluate airflow characteristics. The ultimate goal is to leverage these insights for the design and optimization of a novel pediatric endotracheal tube that is anatomically informed and functionally validated.

Originality of the Project:

This research integrates digital twin-based modeling, statistical shape optimization, and flow simulations to address a critical unmet need in pediatric airway management, providing innovative solutions beyond existing commercial devices.

			
Expected Outcomes of the Project (Targets):			
An in-silico tested digital airway model, post process flow results			
Expected skills and abilities that student(s) gain at the end of the project:			
Image processing, segmentation, statistical shape analysis, digital twin modeling, and biomedical device design.			
Number and program of students requested: (if flexible, indicate so, e.g. 2-4)	2 (Biomedical Engineering; flexible if needed)		
Recommended Background for students:			
Medical imaging, Biomechanics, Matlab, CAD, ML			